Endalkachew Y. Gebru

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EDUCATION

University of Minnesota – Twin Cities *Minneapolis, MN*
Electrical and Computer Engineering, PhD *Expected: 2028*

- CGPA: 3.98 | Advisor: Dr. Sachin S. Sapatnekar
- Relevant Coursework: Analog IC Design, VLSI & VLSI Design Automation, Analysis of Numerical Methods, Design Automation Techniques for Variation-Aware Computing, Machine Learning, Deep Learning, RF/Microwave Circuits Design, Wireless Hardware System Design, Heterogeneous Integrated Circuits, Matrix Theory.

Addis Ababa University *Addis Ababa, Ethiopia*
Electrical and Computer Engineering, BSc *May 2022*

- GPA: 4.0 | Thesis: Long-Range Visible Light Communication System Design and Implementation
- Relevant Coursework: Analog and Digital Electronics, Computer Architecture, Control Theory, EM Waves and Guided Structures, Data Structures and Algorithms, Mobile and Wireless Communication

WORK EXPERIENCE

Analog EDA R&D | Cadence Design Systems *(San Jose, CA, USA)* *May 2025 – Aug 2025*

- Worked on analog circuit and layout design automation and optimization using machine learning techniques

Graduate Research Assistant | University of Minnesota *(Minnesota, USA)* *Aug 2023 – present*

- Working on analog circuit and layout design automation and optimization with ML techniques

Research Assistant | McGill University *(Montreal, Canada | Remote, full-time)* *Jun 2022 – Aug 2023*

- Developed a MATLAB-based wind engineering and uncertainty quantification framework
- Worked on PC-NARX modeling of stochastic wind response and a Tuned Mass Damper (TMD) simulator

PUBLICATIONS & MANUSCRIPTS

[1] **E. Y. Gebru**, S. Ghosh, R. Harjani, and S. S. Sapatnekar, "Parasitics-Aware Framework for Integrated OTA Sizing and Layout Synthesis in FinFET Technologies," in *Proc. Int. Symp. Quality Electronic Design (ISQED)*, 2026.

[2] S. Ghosh, **E. Y. Gebru**, C. V. Kashyap, R. Harjani, and S. S. Sapatnekar, "Accelerating OTA Circuit Design: Transistor Sizing Based on a Transformer Model and Precomputed Lookup Tables," in *Proc. Design, Automation & Test in Europe (DATE)*, 2025.

[3] **E. Y. Gebru**, R. Harjani, and S. S. Sapatnekar, "Co-MiX: A Co-ordinated Multi-Agent Framework for MiXed-Signal IC Design from Intent to GDSII," *Submitted to DATE 2027*.

[4] S. Ghosh, S. S. Peri, Ramprasath S., S. A. Berhan, **E. Y. Gebru**, R. Harjani, and S. S. Sapatnekar, "COmPOSER: Circuit Optimization of mm-wave/RF circuits with Performance-Oriented Synthesis for Efficient Realizations," *Submitted to DAC 2026*; arXiv:2603.20486.

[5] **E. Y. Gebru**, Ramprasath S., R. Harjani, and S. S. Sapatnekar, "Row-Based Layout Synthesis for Analog Circuits Using Height-Quantized Primitives," *Submitted to IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems (TCAD)*.

PROJECTS

- **AI Workload Mapping on Heterogeneous Integrated Circuits**
  Optimizing AI workload mapping on heterogeneous integrated circuits using HISIM
- **CMOS Readout IC (ROIC) Layout Automation** (Collaboration with Nanohmics Inc.)
  Automated layout generation for an infrared bolometer readout IC using SkyWater 130 nm technology and ALIGN
- **Fully Differential CMOS Biquad Filter**
  Design and implementation of a low-power biquad filter suitable for hearing aids using 65 nm technology
- **FastPlace Algorithm Implementation**
  Developed and tested the algorithm on IBM benchmark circuits
- **Static Timing Analysis (STA)**
  Created a C++ framework to compute total delay, slack, and critical paths of circuits using NLDM models
- **Simulated Annealing-Based Floorplanning**
  Built a simulated annealing engine to optimize floorplanning for reduced layout area and wire length
- **14-bit General-Purpose Microprocessor**
  Designed a microprocessor with microprogrammed control and an 18-instruction set; implemented in Proteus

TECHNICAL SKILLS

Programming: MATLAB, Python, Java, C++
EDA/Simulation Tools: Cadence Virtuoso and SKILL, Magic, LTspice, Ngspice, Simulink
ML Libraries: PyTorch, Scikit-Learn

ACHIEVEMENTS

- **2022 AAU Gold Medal Award:** awarded for academic excellence to the top-ranking undergraduate in a bachelor's program
- **POSCO African Fellowship award (2020–2022):** three-year financial award for outstanding academic performance